Chapter 5 solutions

CH05-E1

1. Cofactors:

$$C_{11} = 1 - 3 = -2,$$
$$C_{12} = -(0 - 6) = 6.$$

The third entry of row 1 is zero, so

$$\det A = 1(-2) + 2(6) = 10.$$

2. Adding twice column 1 to column 2 gives

$$B = \begin{bmatrix} 1 & 4 & 0 \\ 0 & 1 & 3 \\ 2 & 5 & 1 \end{bmatrix}.$$

Expansion along row 1 gives

$$\det B = 1(1 - 15) - 4(0 - 6) = 10.$$

3. Write $B = AE$, where the elementary column-addition matrix satisfies $\det E = 1$. Therefore

$$\det B = \det A \det E = \det A.$$

Geometrically, the shear preserves area or volume.

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CH05-E2

1. The six terms are

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$
$$- a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

2. Column orders

$$(1,2,3), \quad (2,3,1), \quad (3,1,2)$$

are even. The odd permutations are $3,2,1$, $1,3,2$, and $2,1,3$.

3. Take

$$M = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 4 \end{bmatrix}.$$

Only

$$2 \cdot 3 \cdot 4 = 24$$

and

$$1 \cdot 1 \cdot 1 = 1$$

are nonzero. Both have even sign, giving

$$\det M = 25.$$

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CH05-E3

1.

$$C_{11} = 10, \qquad C_{12} = 4, \qquad C_{13} = -2.$$

2.

$$\det A = 2(10) + 1(4) + 0(-2) = 24.$$

3. The row 2 cofactors are

$$C_{21} = -(4) = -4,$$
$$C_{22} = 8, \qquad C_{23} = -(4) = -4.$$

Hence

$$\det A = (-1)(-4) + 3(8) + 1(-4) = 24.$$

CH05-E4

1. The determinant is 10, and the cofactor matrix is

$$C = \begin{bmatrix} -2 & 6 & -2 \\ -2 & 1 & 3 \\ 6 & -3 & 1 \end{bmatrix}.$$

$$\operatorname{adj} A = C^T = \begin{bmatrix} -2 & -2 & 6 \\ 6 & 1 & -3 \\ -2 & 3 & 1 \end{bmatrix}.$$

$$A^{-1} = \frac{1}{10} \begin{bmatrix} -2 & -2 & 6 \\ 6 & 1 & -3 \\ -2 & 3 & 1 \end{bmatrix}.$$

2. Multiplication gives diagonal entries 10 and off-diagonal entries 0.

$$A \operatorname{adj} A = 10I.$$

3. For $Ax = b$,

$$x = \frac{1}{\det A} \operatorname{adj}(A) b.$$

This is exact but expensive at large size. The adjugate has n^2 minors; use pivoted factorization for large numerical systems.

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CH05-E5

1.

$$\det A = -2, \qquad \det A_1 = -4, \qquad \det A_2 = 2.$$

$$x_1 = 2, \qquad x_2 = -1.$$

2. Verification:

$$3(2) + 4(-1) = 2, \qquad 5(2) + 6(-1) = 4.$$

3. For $b = (2, 5)^T$,

$$\det A_1 = -8, \qquad \det A_2 = 5,$$

$$x_1 = 4, \qquad x_2 = -\frac{5}{2}.$$

CH05-E6

1.

$$P = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \qquad \det P = 5.$$

The counterclockwise ordering gives positive orientation, so signed and unsigned areas are both 5.

2. Define

$$X = w_2 z_3 - w_3 z_2,$$
$$Y = w_3 z_1 - w_1 z_3, \qquad Z = w_1 z_2 - w_2 z_1.$$

Direct expansion gives

$$w_1 X + w_2 Y + w_3 Z = 0,$$

because the terms cancel in pairs. The same expansion proves $z \cdot (w \times z) = 0$.

3.

$$w \times z = (-2, 1, 1)^T.$$

Triple product identity:

$$\det[w\ z\ w \times z] = \|w \times z\|^2 = 6.$$

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CH05-E7

1. Factor c from each of the n columns.

$$\det(cA) = c^n \det A.$$

2.

$$1 = \det(Q^T Q) = (\det Q)^2,$$

Hence

$$\det Q = \pm 1, \qquad |\det Q| = 1.$$

3. If $S^T = -S$ and n is odd,

$$\det S = \det(S^T) = \det(-S) = (-1)^n \det S = -\det S.$$

Thus $\det S = 0$, proving singularity.

If n is even, the sign is positive and this argument imposes no zero determinant. For example,

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

has determinant 1.

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CH05-E8

1. Expanding along row 1 gives

$$M_{11} = \begin{bmatrix} d & 1 \\ 2 & a+d \end{bmatrix},$$
$$\det M_{11} = d(a+d) - 2.$$
$$M_{12} = \begin{bmatrix} c & 1 \\ 0 & a+d \end{bmatrix},$$
$$\det M_{12} = c(a+d).$$

Since the third entry is zero,

$$\det A = a[d(a+d) - 2] - bc(a+d).$$

2. Choose $a = d = 0, b = c = 1$, giving

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}.$$

Columns 1 and 3 both equal $(0, 1, 0)^T$, so rank is below 3 and the determinant is zero. The columns lie in a plane, and their three-dimensional volume vanishes.

3. $a = d = 1, b = c = 0$:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix}.$$
$$\det A = 1(2-2) = 0.$$

The characteristic polynomial is $\det(\lambda I - A) = \lambda(\lambda - 1)(\lambda - 3)$, giving eigenvalues $0, 1, 3$. Their sum is $4 = \operatorname{tr} A$ and product $0 = \det A$. A nonzero trace therefore does not imply invertibility.

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CH05-E9

$A^{-1} = \operatorname{diag}(1, 1/2)$, so $CA^{-1} = I_2$ and $CA^{-1}B = I_2$. Hence

$$S = D - I_2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \det S = 1.$$

Choose $L = \begin{bmatrix} I_2 & 0 \\ -I_2 & I_2 \end{bmatrix}$. Then $\det L = 1$ and $LM = \begin{bmatrix} A & I_2 \\ 0 & S \end{bmatrix}$, giving $\det M = \det A \det S = 2$.

Singularity of A only makes A^{-1} unavailable in that formula; it does not force M to be singular. Partition $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ into scalar blocks: $A = 0$ is singular, yet the whole determinant is -1.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 5.

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